

The solver does not deliver the β_{sub0} it is given

Diagnosis and remediation plan for `lunar_regolith_DSM`

Working note — evidence from batch 2026-08-07__07.26.38_wedge_bc

2026-08-25

Summary. The realized kinetic coefficient is $1.22\times$ the requested `-beta_sub0`, so every run sublimates and deposits 18 % slower than intended. The capillary length d_0 is exact. The excess is precisely the two Karma–Plapp thin-interface corrections that `lunar_main.c:829` adds to τ_{sub} for the asymptotics to subtract back off — and which the measurement says are never subtracted. One of the two (the thermal one, 13.1 of the 17.8 points) is wrong on its face and should be fixed regardless. The fix is to τ_{sub} , not to the Gibbs–Thomson prediction, which is correct as written.

1. What was measured

The wedge geometry places an apex-centered ice annulus in a tapering pore with the virtual apex at $y = L_y/2$, so on the centerline the interface normal is *exactly* $\pm\hat{x}$ and v_n is a one-dimensional displacement rate. `postprocess/wedge_gt_velocity.py` measures it on both menisci of all nine runs and compares against

$$v_n = \frac{\sigma \rho_{vs}(T) / \rho_{vs}(T_0) - d_{0,\text{sub0}} \chi}{\beta_{\text{sub0}}}, \quad \sigma = \frac{\rho_v - \rho_{vs}(T)}{\rho_{vs}(T)}. \quad (1)$$

The solver recovers 0.826 (inner) / 0.811 (outer) of that rate — flat over 90 days, and flat across nine vapor-boundary combinations whose residual driving force $\sigma - d_0\chi$ spans two orders of magnitude. Interface-extraction noise is 0.5 % against an 18 % effect, and three independent v_n estimators agree on the median to 1 %, so the shortfall is not a measurement artifact.

2. The regression: which coefficient is wrong

Rather than assume Eq. (1) and read off a ratio, fit both coefficients freely. Every snapshot supplies a triple (σ, v_n, χ) , so

$$\sigma = \beta v_n + d_0 \chi \quad (\text{no intercept}) \quad (2)$$

is an ordinary two-column least-squares problem. **The design is identifiable because the two menisci carry opposite-sign curvature** — the inner one is concave, the outer convex — while the 3×3 reservoir grid moves v_n at nearly fixed χ . Those are two genuinely different directions:

χ spans	$-5.259 \times 10^3 \dots +3.459 \times 10^3 \text{ m}^{-1}$
v_n spans	$-1.459 \times 10^{-12} \dots +1.378 \times 10^{-12} \text{ m/s}$
$\text{corr}(v_n, \chi)$	-0.7095 (correlated, but far from collinear)
condition number	2.58

Fitting $n = 3492$ points (nine runs $\times$ two fronts $\times$ every fifth snapshot after the initial transient):

	β [s/m]	$\times \beta_{\text{sub0}}$	d_0 [m]	$\times d_{0,\text{sub0}}$
pooled, both fronts	$7.2324 \times 10^5 \pm 1.8 \times 10^2$	1.2214	$1.0172 \times 10^{-9} \pm 3.5 \times 10^{-14}$	1.0006
left front only	$7.1384 \times 10^5 \pm 1.6 \times 10^2$	1.2055	$1.0166 \times 10^{-9} \pm 2.7 \times 10^{-14}$	1.0000
right front only	$7.2716 \times 10^5 \pm 1.3 \times 10^2$	1.2280	$1.0167 \times 10^{-9} \pm 3.5 \times 10^{-14}$	1.0001

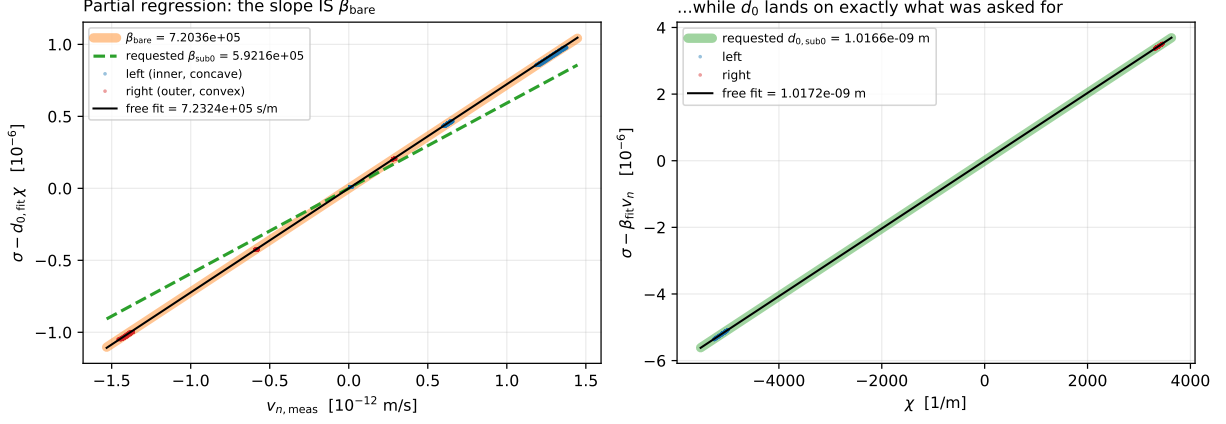


Figure 1: Partial-regression views of the same fit. **Left:** with the fitted curvature term removed, $\sigma - d_0\chi$ against v_n is a straight line through the origin whose slope is the realized β . The black fit lies inside the β_{bare} band (orange) and is visibly steeper than the requested β_{sub0} (green dashed). **Right:** with the kinetic term removed, the slope against χ falls inside the $d_{0,\text{sub0}}$ band — the capillary length is delivered exactly. Both fronts, all nine runs, every fifth snapshot.

$R^2 = 0.999997$ pooled; the rms residual is 5.4×10^{-9} , i.e. 0.14% of $|\sigma|$. Holding $d_{0,\text{sub0}}$ fixed and fitting β alone gives 7.2106×10^5 , within 0.3% of the free fit — the two parameters are not trading off against each other.

Robustness. Leave-one-run-out (drop each of the nine in turn and refit) moves β over 7.2204 – 7.2419×10^5 , a spread of 0.30% of the mean. No single vapor-BC condition drives the result. The residual left/right split (1.206 vs 1.228) is $\pm 1\%$ and is consistent with the quoted σ -extraction sensitivity (0.7% left, 0.3% right).

Conclusion. $d_{0,\text{sub0}}$ is recovered to 0.06%. The Gibbs–Thomson *curvature* physics is exact and the prediction formula is right. The entire deficit lives in β , and it is a clean multiplicative factor of 1.22.

3. β lands on the *bare* kinetic coefficient

Dividing the phase residual by $3 \text{ mob}/\epsilon$ gives the Moure & Fu form

$$\tau_{\text{sub}} \partial_t \phi = \epsilon^2 \nabla^2 \phi - f'(\phi) + \lambda_{\text{sub}} g(\phi) \frac{\rho_{vs}(T)}{\rho_{\text{ice}}} \sigma, \quad (3)$$

whose sharp-interface limit is exactly the Gibbs–Thomson condition Eq. (1), with the two coefficients it realizes given by

$$d_{0,\text{realized}} = d_{0,\text{sub0}}, \quad \beta_{\text{realized}} = \underbrace{\frac{\tau_{\text{sub}} d_{0,\text{sub0}}}{\epsilon^2}}_{\beta_{\text{bare}}} - C_{\text{corr}}. \quad (4)$$

C_{corr} is an $O(\epsilon)$ term generated by σ varying *inside* the diffuse interface. `lunar_main.c:829` pre-inflates τ_{sub} by exactly what it assumes C_{corr} to be, so that the two cancel and $\beta_{\text{realized}} = \beta_{\text{sub0}}$:

$$\tau_{\text{sub}} = \epsilon \lambda_{\text{sub}} \left(\underbrace{\frac{\beta_{\text{sub}}}{a_1}}_{82.2\%} + \underbrace{\frac{a_2 \epsilon}{\text{diff_sub}}}_{13.1\%} + \underbrace{\frac{a_2 \epsilon}{\text{dif_vap}}}_{4.7\%} \right), \quad \beta_{\text{bare}} = 7.2036 \times 10^5 \text{ s/m}. \quad (5)$$

(The bracket is written in the $\rho_{vs}/\rho_{\text{ice}}$ -scaled variables the code carries internally; β_{bare} above is already converted back to the unscaled units of β_{sub0} .) The fit gives $\beta/\beta_{\text{bare}} = 1.0040$. $C_{\text{corr}} \approx 0$:

nothing is subtracted, so the inflation survives intact, and $\beta_{\text{bare}}/\beta_{\text{sub0}} = 1.2165 \Rightarrow$ ratio 0.8220 against a measured 0.826/0.811.

This is a defect in a *model parameter definition*. Eq. (1) is the target physics and is correct; the solver simply does not realize the β it was handed.

4. Why neither correction materialises

4.1 The thermal term is wrong on its face

`a2*eps/diff_sub` treats latent heat as if it drove the phase field at unit strength. It does not: temperature reaches σ only through $\rho_{vs}(T)$, so that channel carries a Clausius–Clapeyron factor. Working the inner balance per channel and reading off $d\sigma/d(v_n\epsilon)$, in the same scaled variables as Eq. (5):

channel	correct coefficient	value
vapor	$1/D_v$	5.27×10^4
	<i>coded:</i> <code>1/dif_vap</code>	4.59×10^4 ✓
thermal	$(d\rho_{vs}/dT) L_{\text{sub}}/k$	$1.0 \times 10^2 (k_{\text{ice}}) - 1.2 \times 10^4 (k_{\text{air}})$
	<i>coded:</i> <code>1/diff_sub</code>	1.29×10^5 ×

Over-weighted by $11\times$ to $1300\times$ depending on where in the interface k is evaluated — and `ThermalCond` interpolates between the two, so *any* reasonable choice leaves this term negligible. This is 13.1 of the 17.8 points and is a defect independent of anything else in this note.

4.2 The vapor term is right, and still is not delivered

It is worth being explicit that ξ_v is *not* implicated here. In the inner region ξ_v multiplies the vapor diffusion and the ice source identically, so it cancels between them and survives only in storage/diffusion, $v_n\epsilon/(\xi_v D_v) = 2.4 \times 10^{-11}$. The unscaled `dif_vap` in Eq. (5) is correct; substituting $\xi_v \text{dif_vap}$ would put the bracket at 6.36 instead of 0.133 and predict a $58\times$ shortfall rather than $1.22\times$.

So the vapor term is structurally sound, yet the measurement says its 4.7 points are not subtracted either. **Leading hypothesis, not established:** vapor diffusion is one-sided through `phi_aef`, and the $-a_1 a_2 \epsilon/D$ correction is derived for the symmetric model. The one-sided case requires an anti-trapping current (Karma 2001) to remove solute-trapping and interface-stretching terms before a clean correction of that form emerges; this solver has no such current, so the symmetric-model correction has no reason to apply.

5. What this has been costing

β_{sub0} encodes the condensation coefficient, $\beta \propto 1/\alpha_c$. A realized β that is $1.2165\times$ too large is an α_c that is $0.822\times$ too small:

requested <code>-beta_sub0</code>	$= 5.9216 \times 10^5$	$\alpha_c = 1.341 \times 10^{-2}$
realized β_{eff}	$= 7.2036 \times 10^5$	$\alpha_c^{\text{eff}} = 1.102 \times 10^{-2}$ (82.2%)

Worse, because the inflation is *additive*, $\beta_{\text{eff}} = \beta_{\text{sub0}} + \Delta$ with $\Delta = a_1 a_2 \epsilon (1/D_T + 1/D_v) \rho_{\text{ice}}/\rho_{vs}(T)$, so Δ is a **hard floor on the kinetics the model can represent at all** — no value of `-beta_sub0`

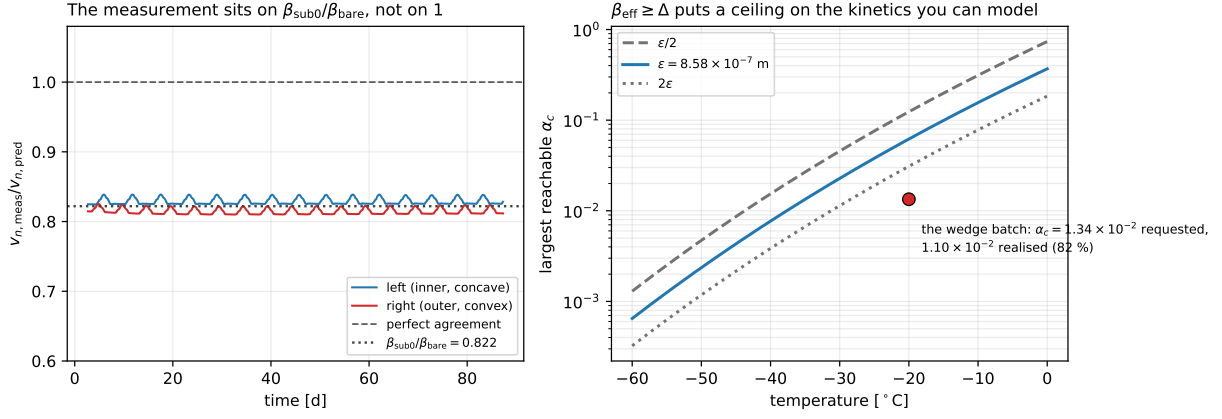


Figure 2: **Left:** the measured ratio tracks $\beta_{\text{sub0}}/\beta_{\text{bare}}$ for the whole run on both fronts. **Right:** the ceiling on the condensation coefficient implied by $\beta_{\text{eff}} \geq \Delta$, as a function of temperature and ϵ .

can go below it. $\Delta \propto \epsilon$ and $\propto 1/\rho_{vs}(T)$, so coarser interfaces and colder temperatures are both worse:

T [°C]	ρ_{vs} [kg/m ³]	Δ [s/m]	largest reachable α_c
-60	8.885×10^{-6}	1.225×10^7	0.0006
-40	1.055×10^{-4}	1.031×10^6	0.0077
-30	3.123×10^{-4}	3.484×10^5	0.0228
-20	8.487×10^{-4}	1.282×10^5	0.0619
-10	2.139×10^{-3}	5.086×10^4	0.1561
0	5.047×10^{-3}	2.156×10^4	0.3684

(at $\epsilon = 8.584 \times 10^{-7}$ m; halving ϵ halves every Δ .) At -60°C the floor is $20.7\times$ the reference β : the model would be running kinetics roughly twenty times slower than requested no matter what is asked for. **Any earlier result whose conclusion depends on the absolute rate — rather than on a scaling exponent or a ratio — should be re-examined against this factor.**

6. What has been ruled out

- **Grid pinning / lattice friction.** The deficit is a constant *factor*, identical on fronts whose speeds differ $60\times$. A threshold or friction cannot behave that way.
- **Interface-extraction bias.** Three independent v_n estimators agree on the median to 1%; the residual scatter is 0.5%, zero-mean, and locked to sub-cell phase.
- **A mis-converted a_1 or a_2 .** ϵ here is Moure & Fu’s ϵ , and is deliberately *not* Kaempfer & Plapp’s W ($W_{\text{KP}} = \sqrt{2}\epsilon$); the constants already absorb the difference. Mapping to K&P’s $[-1, 1]$ form gives $d_{0,\text{sub}} = a_1^{\text{KP}}\sqrt{2} \cdot 4 \cdot \epsilon/\lambda_{\text{sub}} = 5.0000\epsilon/\lambda_{\text{sub}}$ against the coded $a_1 = 5.0$, and $a_1^{\text{KP}}a_2^{\text{KP}}\sqrt{2} = 0.78338$ against the coded $a_1a_2 = 0.7905$ (0.91%). This is already documented at `enceladus_DSM/src/assembly.c:46-58`. The free fit recovering $d_{0,\text{sub0}}$ to 0.06% is an independent measurement of a_1 .
- ξ_v . Cancels from the inner balance (§4).

7. Remediation plan

7.1 Step 1 — measure before changing anything: the β_{sub0} ladder

Everything above rests on one $(\epsilon, \beta_{\text{sub0}}, T)$ point. Because the inflation is additive and β_{sub0} is not, the predicted shortfall is a strong function of β_{sub0} — so a ladder in β_{sub0} separates the hypotheses without touching the solver. Four runs, identical but for `-beta_sub0`, with `-t_final` and `-dtmax` set per arm so all four take the same 3406 steps and move the menisci the same ~ 2 control-point spacings:

$\times$	<code>-beta_sub0</code>	predicted ratio	τ_{sub} [s]	<code>-dtmax</code>	<code>-t_final</code>
0.10	5.9216e4	0.3160	135.8	1.485e2	5.0577e5 (5.9 d)
0.25	1.4804e5	0.5359	200.2	2.188e2	7.4548e5 (8.6 d)
1.00	5.9216e5	0.8220	522.1	5.707e2	1.9440e6 (22.5 d)
4.00	2.3686e6	0.9487	1809.8	1.978e3	6.7381e6 (78.0 d)

```
./scripts/HPC/submit_batch.sh --tag beta_ladder \
--tests-file scripts/HPC/tests_beta_ladder.txt
```

Read `meas` / `pred` from each run's `wedge_gt_velocity.py` summary:

- 0.316/0.536/0.822/0.949 — $C_{\text{corr}} = 0$ confirmed. Proceed to Step 2.
- 1.00 at all four — the corrections do operate; the post-processing is wrong and this note is wrong with it.
- 0.82 at all four — a constant factor from some other source; start over.

An intermediate curve — following the shape but not the amplitude — would mean C_{corr} is real but mis-sized, which is the outcome that most needs the Step-4 modeling work rather than a coefficient change.

7.2 Step 2 — the τ_{sub} change

Make the correction explicit and switchable rather than silently baked in, so both forms stay runnable and the ladder can be repeated against either. In `lunar_main.c` around line 829:

```
/* Thin-interface corrections. Each channel enters with the strength
 * with which it actually drives u = (rho_v - rho_vs(T))/rho_ice, NOT
 * with unit strength. See docs/gt_deficit/. */
PetscReal c_vap = a2 * eps / user.dif_vap;
PetscReal c_therm = a2 * eps * drhovs_dT * user.lat_sub / k_eff;
if (!user.thin_iface_corr) { c_vap = 0.0; c_therm = 0.0; }
tau_sub = eps * lambda_sub * (beta_sub / a1 + c_therm + c_vap);
```

with `-thin_iface_corr` defaulting to whatever the ladder says, `drhovs_dT` from the existing `RhoVS_I` derivative output — `lunar_main.c:735` already requests `d_rhovs` and then never uses it, and it is exactly $d\rho_{\text{vs}}/dT = 8.1521 \times 10^{-5} \text{ kg m}^{-3} \text{ K}^{-1}$ at -20°C , and `k_eff` from `ThermalCond` at $\phi = 1/2$. Print the three bracket terms and $\beta_{\text{bare}}/\beta_{\text{sub0}}$ in the run header next to τ_{sub} , so every future `outp.txt` states how far the realized kinetics sit from the requested ones.

7.3 Step 3 — recalibrate what depends on β

- `preprocess/comp_eps.py` sizes ϵ using `beta_uns (eps_ceiling, kin_over_curv)`. With the correction in place the relation becomes implicit — ϵ sets Δ , which sets β_{eff} , which feeds the ϵ bound — so it needs either an iteration or an explicit note that it reports the *requested* rather than the realized kinetics.
- Re-derive ϵ for the standard -20°C geometries after the change and confirm the mesh element counts do not move; if they do, meshes must be regenerated before any comparison run.
- Audit past conclusions for absolute-rate dependence. Exponent studies and dimensionless ratios are unaffected; anything quoting a time-to-event or an absolute growth rate is off by the factor in §5 for its own (ϵ, T) .

7.4 Step 4 — the open modeling question

If the ladder confirms $C_{\text{corr}} = 0$, the honest position is that this model has *no* thin-interface correction, and dropping both terms is a calibration fix, not a derivation. Making the correction genuinely valid means adding an anti-trapping current to the vapor equation so the one-sided asymptotics close. That is a real project — new flux term in `assembly.c`, matching Jacobian, and re-verification with `-snes_test_jacobian` — and should only be undertaken if a case appears where the $O(\epsilon)$ physics actually matters. At present $\epsilon/r \approx 4 \times 10^{-3}$ and the interface is deeply attachment-limited, so it does not.

7.5 Interim: how to get the kinetics you want today

Until Step 2 lands, pre-compensate. To realize a target β , pass $\beta_{\text{sub0}} = \beta_{\text{target}} - \Delta$; at -20°C and $\epsilon = 8.584 \times 10^{-7}$ m that is 4.6396×10^5 in place of 5.9216×10^5 . This fails outright when $\beta_{\text{target}} < \Delta$ — see the floor table — and it is a workaround, not a fix: it must be recomputed for every (ϵ, T) and it silently stops working when someone changes either.